Journal of Pharmaceutical and Biomedical Analysis 204 (2021) 114251

Contents lists available at ScienceDirect

Journal of Pharmaceutical and Biomedical Analysis

journal homepage: www.elsevier.com/locate/jpba

Rapid and comprehensive monoclonal antibody Characterization using microfluidic CE-MS

Li Cao*, Daniel Fabry, Kevin Lan

CMC Analytical, GlaxoSmithKline, 1250 S. Collegeville Road, UP 1400, Collegeville, PA, 19426, USA

ARTICLE INFO

Article history:
Received 19 April 2021
Received in revised form 29 June 2021
Accepted 1 July 2021
Available online 7 July 2021

Keywords:
monoclonal antibody (mAb)
microfluidic CE-MS
peptide mapping
native intact analysis
post-translation modifications (PTMs)
charge variants

ABSTRACT

The identification and control of monoclonal antibody (mAb) critical quality attributes (CQAs) is a key component of quality by design (QbD). In this work, rapid peptide mapping and native intact charge variants analysis have been developed to comprehensively characterize and monitor mAb CQAs using a microfluidic capillary electrophoresis - mass spectrometry (CE-MS) platform.

The ultrafast peptide mapping simultaneously analyzed multiple CQAs, including protein primary structure, oxidation, deamidation, succinimide, C-terminal lysine (Lys) clipping, N-terminal cyclization, and glycosylation. The microfluidic CE-MS based peptide mapping acquired results comparable to conventional but lengthy liquid chromatography – MS (LC-MS) based approach.

The native intact analysis resolved mAb charge variants with a comparable resolution as commonly achieved using capillary isoelectric focusing (cIEF). Charge variants' identities were assigned based on characteristic mass shifts, knowledge learned from peptide mapping, and changes in electrophoretic mobility. Major mAb glycoforms of each charge variants were resolved and identified in the deconvoluted mass spectra.

Furthermore, a model simulation was performed to reconstruct intact deconvoluted mass spectra using peptide mapping results. The reconstructed and experimentally determined intact deconvoluted mass spectra were highly correlated, suggesting that our data collected at the peptide level and intact level were consistent and highly comparable.

Overall, the microfluidic CE-MS based peptide mapping and native intact charge variants analysis are high-throughput methods that have great potential to support biopharmaceutical development.

1. Introduction

Since the first monoclonal antibody (mAb) was licensed in 1986, mAb therapeutics have become a predominant treatment modality for various diseases, such as oncology, hematology, dermatology, etc [1]. Currently, there are nearly 870 antibodies in clinical development, and the FDA just approved the 100th mAb therapeutic in Apr, 2021 [1]. The mAb market is growing at a fast rate from US $107 billion in 2020 to $180 billion in 2025 [2]. In the meantime, biopharmaceutical companies have gradually moved to a scientific and risk-based quality by design (QbD) approach. One key concept of QbD is to identify critical quality attributes (CQAs) during early drug product and process development [3]. The mAb primary structure, post-translational modifications (PTMs), N-linked glycosylation profile and charge variants are CQAs or potential CQAs. Comprehensive, in-depth characterization and continuous monitoring are critical for CQA identification, control strategy implementation, and quality target product profile (QTPP) completion.

Characterization of biotherapeutics' primary structure, PTMs, and N-linked glycosylation can be done simultaneously at the peptide level by peptide mapping [4], which is a bottom up approach to analyze enzymatically digested peptide mixtures using a separation technique (such as chromatography or electrophoresis) coupled with optical detection or a mass spectrometer (MS) [4]. In addition to the most commonly used reversed phase liquid chromatography (RP-LC), hydrophilic interaction liquid chromatography (HILIC) was also used to separate oxidized and deamidated peptides with their native counterparts [5]. However, all these methods contain a separation step of at least 60 min. This long run time significantly limits peptide mapping applications.

Biotherapeutics charge variants are commonly characterized by capillary isoelectric focusing (cIEF) [6], ion exchange chromatography (IEX) [7], and capillary zone electrophoresis (CZE) [8], with optical detection. However, optical detection allows identification

* Corresponding author.
E-mail address: li.7.cao@gsk.com (L. Cao).

https://doi.org/10.1016/j.jpba.2021.114251
0731-7085/

only for analytes that standard migration times have been determined. In contrast, MS is a highly selective and sensitive analytical technique that provides mass information and elucidates structural properties at a molecular level. Recently, there are emerging activities to hyphenate separation technologies to MS for charge variant analysis [9–11].

ZipChip is a microfluidic capillary electrophoresis (CE) with an integrated nano-electrospray ionization (nano-ESI) source that couples to a mass spectrometer [12]. The interface can be easily installed and removed from a MS inlet within minutes [12]. Integration of nano-ESI source on the microfluidic chip eliminates the dead volume between the separation channel and the ESI source to prevent band broadening. In addition, the microfluidic separation requires lower amounts of sample and can provide higher sensitivity, compared with analytical scale LC. Since the launch of ZipChip, it has been applied for native intact mAb [13–15], mAb-ADC analysis [16], peptide mapping [17], glycan and glycopeptide analysis [18] to support biotherapeutics development.

We developed microfluidic CE-MS based peptide mapping and native intact charge variants analysis methods for mAb CQA characterization and monitoring. The ultrafast 10 min peptide mapping method confirms protein primary structure, identifies and relatively quantifies PTMs, including oxidation, deamidation, succinimide, C-terminal lysine (Lys) clipping, N-terminal cyclization, and N-linked glycosylation profile. The native intact charge variants analysis method requires minimal sample preparation, measures charge variant heterogeneity, and provides their identity in one assay. In this work, GSK-manufactured mAbs were analyzed by peptide mapping and native intact methods using a single microfluidic CE-MS system. The CQA information determined using the microfluidic CE-MS system were comparable with results determined using conventional methods. Furthermore, the peptide mapping and charge variants results are highly correlated, as indicated by model simulations. The presented work demonstrates fast and reproducible microfluidic CE-MS based peptide mapping and native intact analysis methods for biopharmaceutical drug product and process development.

2. Materials and Methods

2.1. Materials and Reagent

The IgG4 mAb and IgG1 mAb that were used for method development were expressed in Chinese hamster ovary cells and manufactured by GSK. ZipChip HR chips, HRN chips, peptide assay kit, and native antibodies kit for peptide mapping and native intact analysis were obtained from 908 device (Boston, MA). Dithiothreitol (DTT) and Sodium iodoacetate (IAA) were purchased from Sigma Aldrich (St. Louis, MO). Trypsin was purchased from Worthington (Lakewood, NJ).

2.2. Microfluidic CE-MS based Peptide mapping

Trypsin digestion: 250 μg of dried mAb sample was reconstituted in 6 M Guanidine hydrochloride buffer at pH 7.5, followed by 20 min reduction using 1 M DTT at ambient temperature. The reduced protein was then alkylated using 1 M IAA for 30 minutes in the dark. Excessive IAA was quenched by addition of 1 M DTT in the mixture. The solution was desalted using a BioSpin column (Bio-Rad) with diammonium hydrogen phosphate and ammonium dihydrogen phosphate digestion buffer at pH 7.5. 2.5 μL of 5 mg/mL trypsin was added in the reduced-alkylated mAb to reach a 1:20 enzyme: substrate ratio and the mixture was incubated at 37°C for 15 minutes. After digestion, 3 μL of 1 M HCl was added to the solution to quench the digestion. The digested samples were diluted using sample diluent in ZipChip Peptide Assay Kit in a 1:10 ratio.

Microfluidic CE-MS analysis: microfluidic CE-MS analysis was performed on a HR chip via ZipChip interface coupled to a Q Exactive Plus mass spectrometry with Biopharm option. Peptide background electrolyte (BGE) from the Peptide Assay Kit and 1 nL digested and diluted mAb sample were loaded into different HR chip wells, separately. CE filed strength was 500 V/cm, and run time was 10 min. The MS capillary temperature was maintained at 200 °C with 2 units of sheath gas. S-lens RF level was set at 50. The MS was operated in positive polarity mode with mass range from 300 – 1800 *m/z*, resolution at 70000, and AGC target of 1E6, followed by a data-dependent higher-energy collision dissociation (HCD) MS/MS (normalized collision energy = 30) of the five most abundant ions.

2.3. Microfluidic CE-MS based native intact analysis

ZipChip Native Antibody kit (sample diluent and BGE), HRN chip, and ZipChip interface coupled to a Q Exactive Plus mass spectrometry with Biopharm option were used for the native intact analysis. mAb proteins were diluted to 1 mg/mL with sample diluent. BGE and 1 nL mAb sample were loaded on an HRN chip placed in the ZipChip interface, separately. The CE separation conditions were: field strength 500 V/cm, pressure assist start time 0.2 min, analysis time 8.5 min. MS acquisition conditions were: positive polarity in high mass range (HMR) mode, scan range 2500 – 8000 m/z, in-source CID 150 eV, 35000 resolution, microscans 3, AGC target 3e6, maximum inject time 50 ms, the sheath gas flow rate 3, the capillary temperature 300 °C and the S-lens RF level 200.

2.4. Data analysis

Raw data from the peptide mapping and intact analysis were processed using Protein Metrics (PMi) Byos software (version 3.9). Intact spectra reconstruction was also performed using PMi Byos.

Peptide mapping analysis: The post-translational modification (PTM) workflow of PMi Byos was used to process peptide mapping raw data. The MS/MS spectra were searched against the mAb amino acid sequence and reversed decoy sequence. Precursor and fragment mass tolerance was 5 ppm and 20 ppm, respectively. Full specific tryptic digestion with 2 miss cleavages were allowed for peptide identification. Carboxymethylation (+58.0055 Da) of Cys was selected as fixed modification, oxidation (+15.9949 Da) of Met and Trp, deamidation (+0.9840 Da) of Asn and Gln, cyclization (-17.0265 Da) of N-terminal Gln, Lys clipping (-128.0950 Da) of C-terminal peptide were selected as dynamic modifications A 50 common biantennary glycan database was also included for glycopeptide identification. In addition, an in-silico peptide list was created for peptide identification based on MS 1 only. Several small di-peptide, tri-peptide, low abundant peptide with PTMs, and some glycopeptides were identified using the in-silico peptide list.

Relative abundance of each PTM was calculated using following equation:

$$\textit{Relative abundance of a PTM} = \frac{\textit{Peak intensity of a peptide with PTM}}{\textit{Sum peak intensity of peptide with PTM and wildtype peptide}} X\,100\%$$

Relative abundance of each glycan species was calculated using following equation:

$$\textit{Relative abundance of a glycan} = \frac{\textit{Peak intensity of a glycopeptide}}{\textit{Sum peak intensity of all glycopeptide identified using MS/MS and MS1 only}} X\,100\%$$

Intact analysis: The intact workflow of PMi Byos was used to process intact raw spectra. Protein mass range was 143000 to 155000 Da. For auto mass picking, the minimum difference between mass peaks was set to 15 Da, and the maximum number of mass peaks was limited to 10. A list of expected glycans was entered/selected from the mass matching tab, and match mass tolerance was set to 10 Da. PMi Intact software allowed automatic peak integration. The quantitation was based on the deconvoluted peak intensities.

Relative abundance of each charge variant was calculated using following equation:

$$\textit{Relative abundance of a charge variant} = \frac{\textit{Peak intensity of a charge variant}}{\textit{Sum peak intensity of all charge variants}} X\,100\%$$

Intact spectrum reconstruction: Individual PTM info and their relative abundance was imported into the reconstruction table of intact workflow to simulate intact mass spectrum.

2.5. LC-MS based Peptide mapping

Tryptic digestion was performed using the procedure described in the CE-MS based peptide mapping section, except tris digestion buffer was used to replace phosphate digestion buffer [19]. Resulted peptides were injected on a Vanquish/Q Exactive plus UHPLC-MS system for reverse phase separation and mass detection [19]. Raw data were also processed using PMi Byos.

2.6. cIEF

Mixture of sample, isoelectric point (pI) marker, ampholyte, and additives were loaded on a cIEF column of iCE3 (Protein Simple, San Jose, California) following manufacturer guidelines. Charge variants were separated based on their pI and detected by a UV detector.

3. Results and discussion

3.1. mAb peptide mapping by microfluidic CE-MS and LC-MS

GSK-manufactured IgG1 and IgG4 mAbs were analyzed using a microfluidic CE-MS based peptide mapping method. A representative base peak intensity (BPI) electropherogram of the IgG 4 mAb tryptic peptides is shown in Fig. 1A. The microfluidic CE-MS based peptide mapping method run time was 10 min; peptides were migrated and separated between ∼3 to 9 min. The total cycle time for analyses, including instrument self-cleaning and preparing for the next injection, was ∼17 min. In comparison, the same mAb sample was also analyzed using a conventional LC-MS based peptide mapping method. The BPI chromatogram of the sample is shown in Fig. 1B. The separation gradient of LC-MS based mapping method was 60 min, plus an additional 30 min for LC column cleaning and re-equilibration. Hence, the run time of a microfluidic CE-MS based peptide mapping was 5 times shorter than a LC-MS based peptide mapping.

3.2. mAb primary structure coverage

GSK IgG4 mAb raw MS data from microfluidic CE-MS based peptide mapping analysis were processed by a Byonic algorithm for peptide identification [20]. Short peptides with no reliable MS2 spectra were identified based on precursor accurate mass. Following peptide identification, peptide mapping showed 100% primary structure coverage of mAb heavy chain (HC) and light chain (LC). Compared to RP-LC, microfluidic CE has distinct advantages in hydrophilic dipeptide, tripeptide, and large hydrophobic peptide

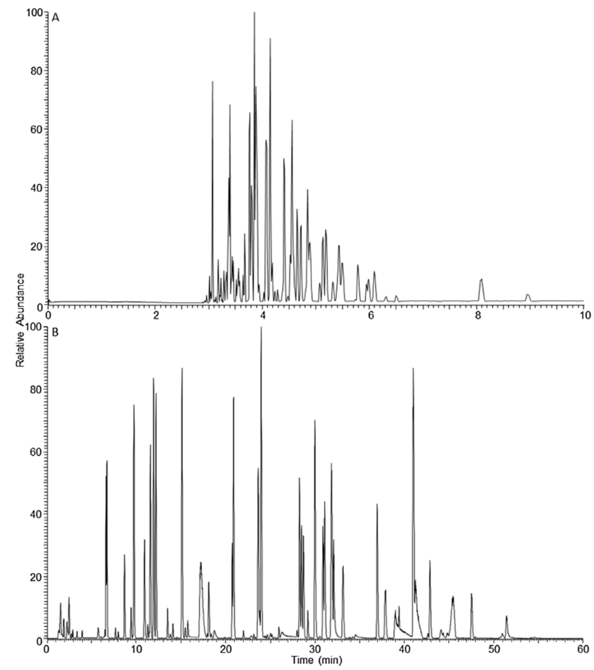

Fig. 1. Comparison of a BPI electropherogram from microfluidic CE-MS based peptide mapping (A) and a BPI chromatogram from LC-MS/MS based peptide mapping (B) of GSK IgG4 mAb.

separation. This IgG4 mAb contains 4 tryptic dipeptides and 4 tryptic tripeptides in its HC and LC. The dipeptides and tripeptides were well separated in the microfluidic CE-MS extracted ion electropherogram (EIE) (Fig. S1A), while most of these peptides coeluted in the early gradient in the LC-MS extracted ion chromatogram (XIC) (Fig. S1B). The two largest tryptic peptides were also selected to compare the separation efficiency between microfluidic CE and RP-LC. Peptide 1 is the largest HC peptide, containing 49 amino acids with a molecular weight of 5060 Da. Peptide 2 is the largest LC peptide, containing 42 amino acids with a molecular weight of 4530 Da. Microfluidic CE-MS EIE showed that the two largest peptides had distinct migration times with sharp, symmetric peaks and good MS intensity (Fig. S2A). In contrast, the same two peptides partially coeluted close to the end of the LC gradient with tailing (Fig. S2B).

3.3. mAb PTM identification and relative quantitation

PTMs of GSK IgG4 mAb, including methionine (Met) oxidation, asparagine (Asn) deamidation and succinimide, HC C-terminal and N-terminal variants etc., were also analyzed. The corresponding wildtype and modified peptide info were listed in Table 1. Identifying and quantifying the relative abundance of deamidated species is challenging because the deamidated peptide is only 0.9840 Da heavier than the native counterpart [21]; therefore, it is critical to chromatographically or electropherographically separate the deamidated peptide from its native peptide [5]. Otherwise, mass spectra of deamidated peptide will overlap with ^{13}C isotopes of its corresponding unmodified peptide. RP-LC often doesn't provide sufficient separation between native and deamidated peptides; therefore, other types of LC separation, like HILIC separation or multi-dimensional LC separation have been developed by others to enhance separation [5,22]. Mass spectrometer data processing software also developed an algorithm to minimize interference of native peptide XIC to deamidated peptide XIC. In contrast, EIEs of a representative Asn deamidated peptide and the native peptide showed that the two peptides were baseline separated, and deamidated peptide migrated later than the native peptide

Table 1
IgG4 mAb CQA related wildtype and modified peptides information.

Peptide	CQA name	Migration time (T, min)	full width half maximum (FWHM, min)	Resolution[a]
XXXXXXXXXXXXXN[Deamid]GK	N1 deamid	3.91	0.02	4.4
XXXXXXXXXXXXXN[succ]GK	N1 succ	3.78	0.02	0.6
XXXXXXXXXXXXXNGK	/	3.76	0.02	/
DTLM[Oxid]ISR	M1 oxid	4.14	0.02	3.1
DTLMISR	/	4.06	0.01	/
XXXXXXXXXXXM[Oxid]XXXXXXXXXXX	M2 oxid	3.43	0.02	0.9
XXXXXXXXXXXMXXXXXXXXXXX	/	3.40	0.02	/
XXXXXXXXXXXXXN[Deamid]XXXXXXXX	N2 deamid	6.14	0.05	1.6
XXXXXXXXXXXXXNXXXXXXXX	/	5.99	0.06	/
XXXXXXG	Lys clip	6.09	0.05	25.8
XXXXXXGK	/	3.9	0.05	/
Q[cyclized]XXXXXXXXXXX	N term cyclized	8.09	0.07	28.7
QXXXXXXXXXXX	/	4.68	0.07	/

[a] Resolution is calculated using equation: $resolution = \mathrm{ABS}[1.18^* \frac{(T1-T2)}{(FWHM1+FWHM2)}]$, 1 represents modified peptide, and 2 represent wildtype peptide.

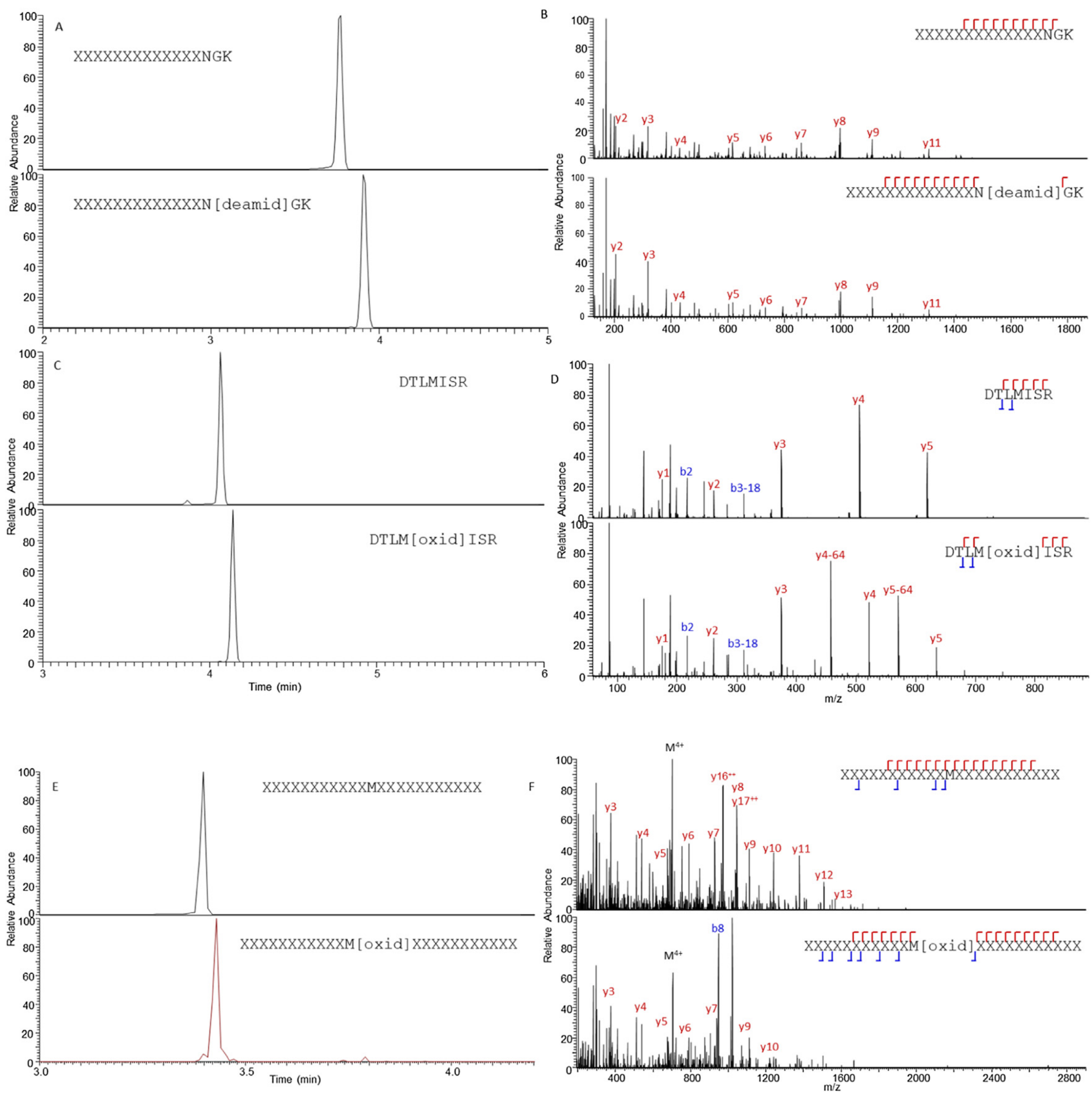

Fig. 2. EIEs and MS/MS spectra of a tryptically digested GSK IgG4 mAb. A, C, and E. EIE of native peptide (top panel) and peptide with deamidation or oxidation (bottom panel). B, D, and F. Representative MS/MS spectrum of the native peptide (top panel) and peptide with deamidation or oxidation (bottom panel). Major y ions, b ions, and charged precursor ions were labeled in the spectra, as well as in the anonymized sequence.

(Fig. 2A), which is consistent with recently published work [17]. The Asn deamidation site was determined by high quality MS/MS spectrum of deamidated and the corresponding native peptide (Fig. 2B).

Succinimide is the intermediate product of Asn deamidation. The IgG4 mAb contains one succinimide modification (N1 succ), which is categorized as a potential CQA. The migration time of the succinimide was the same as its native peptide (EIEs were not

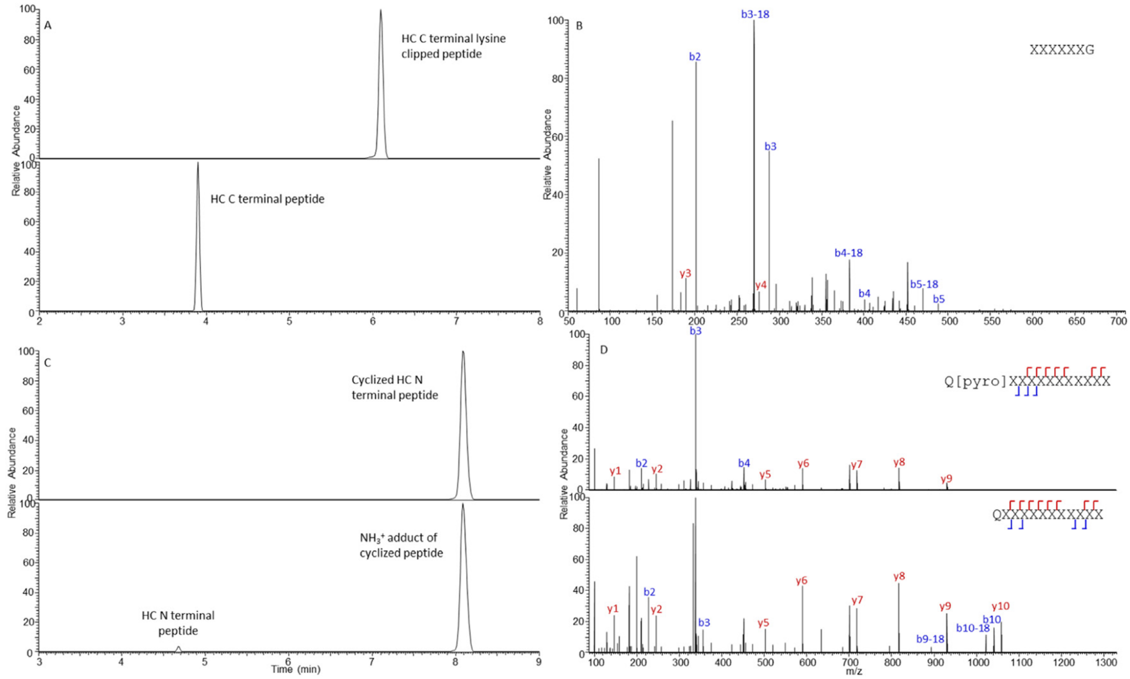

Fig. 3. EIEs and MS/MS spectra of a tryptically digested GSK IgG4 mAb. A and C. EIE of HC N-terminal and C-terminal variants. B and D. Representative MS/MS spectrum of the HC C-terminal Lys clipped peptide and HC N-terminal cyclized peptide.

shown). However, MS is capable of separating the succinimide and native peptide signal, since the succinimide peptide is 17.0265 Da lower than the native peptide. Therefore, relative abundance of succinimide could be calculated accurately.

EIEs of two Met oxidized peptides and corresponding native peptides showed that the oxidized peptide migrated later than the native peptide, which is consistent with recent findings [17]. The peptide pair with lower molecular weight was fully separated (Fig. 2C), and the other peptide pair with higher molecular weight was not fully separated (Fig. 2E). Met oxidation can induce conformational changes, hence impacting electrophoretic mobility of the modified peptide. Microfluidic CE separation has disadvantages and advantages compared with RP-LC separation with regards to Met oxidized peptide separation. A low percentage of Met in source oxidation is commonly observed during electrospray ionization [23], which inherently has the same migration or elution time as the native peptide. Therefore, the Met oxidation relative abundance calculation has interference from the artificial in-source Met oxidation if complete separation of oxidized and unmodified native peptide was not achieved. On the contrary, Met oxidized peptides normally eluted minutes earlier than the native counterpart in RP-LC due to hydrophobicity change. The in-source oxidized peptide usually was well separated with peptide oxidized post-translationally. The calculation using the well-separated oxidized peptide pair showed that the in-source oxidized peptide peak intensity was about 0.05% of the native peptide peak intensity; hence, in-source oxidation interference was minor for the not-well-separated microfluidic CE-MS peptide pair. The advantage of microfluidic CE is that no on-column Met oxidation was observed as in RP-LC, which could be ten times higher than real oxidation when an aged column is used [24]. The Met oxidation site in both oxidized peptides were clearly marked by high quality fragment spectra of the oxidized deamidated and corresponding native peptide (Fig. 2D, F).

EIEs of HC C-terminal peptide variants showed that HC C-terminal Lys clipped peptide migrated about 2 min later than the HC C-terminal peptide with Lys (Fig. 3A). Lys clipping results in a loss of positive charge and a decrease in electrophoretic mobility. MS/MS spectrum of Lys clipped peptide was collected, and the amino acid sequence of the peptide was confirmed (Fig. 3B). The HC C-terminal peptide with Lys was identified based on MS 1 accurate mass only (no fragment spectra was collected) as this peptide is a minority variant and low abundance. EIEs of HC N- terminal variants showed that the cyclized HC N-terminal peptide migrated at about 3.2 min later than the native HC N-terminal peptide (Fig. 3C), since glutamine (Gln) cyclization results in positive charge loss. EIE of HC N-terminal peptide m/z showed two peaks. The peak migrated at ~4.6 min was identified as HC N-terminal peptide, which was confirmed by MS/MS spectra (Fig. 3D). The other peak that had the same migration time with the cyclized HC N-terminal peptide should be an ammonium adduct of the cyclized HC N-terminal peptide, since ammonium acetate was the major component of BGE and sample diluent (Fig. 3C).

3.4. mAb N-linked glycosylation profile analysis

CE-MS is an emerging technology for glycan and glycopeptide analysis [18,25,26], since it combines strengths of high peak capacity of CE and precise measurement of glycan or glycopeptide mass. For GSK mAb IgG4, the N-linked glycosylated peptides of two major glycan species and four minor glycan species were characterized. The glycans attached to these glycosylated peptides are biantennary complex or high-mannose glycans, as confirmed by a separate 2-AB labelled release glycan method. EIEs of these glycosylated peptides showed that major glycan 1 and 2, minor glycan 1, 2, and 3 were partially separated between 6 to 7 min (Fig. S3). Minor glycan 4 is a sum of sialylated glycans migrated between 9 to 10 min (Fig. S3). Glycopeptides with major glycan 1, 2, and minor glycan 3 had tandem mass spectra, which mostly contained signature glycan oxonium ions and fragments of the peptide backbone plus different monosaccharide cleavages; a few y and b ions from the peptide backbone were also detected (data not shown). Glycopeptides with minor glycan 1, 2, and 4 were identified according to precursor accurate mass.

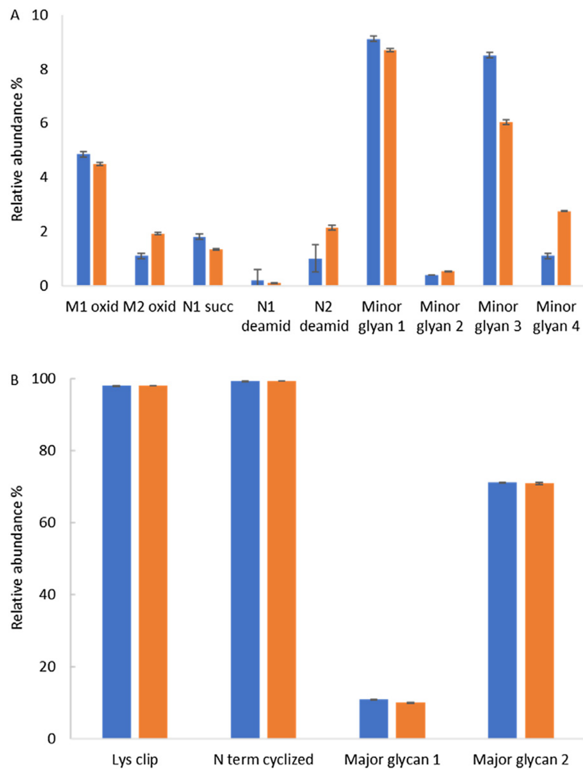

Fig. 4. Relative abundance comparison of selected CQAs of a GSK IgG4 mAb. Results determined using microfluidic CE-MS based peptide mapping (blue) and LC-MS/MS based peptide mapping (orange). Error bars represent the standard deviation for each measured CQA over 6 replicates.

In summary, microfluidic CE-MS based peptide mapping is able to characterize primary structure, multiple PTMs, and glycopeptides in a single run in a high throughput manner. The relative abundance of identified PTMs and glycosylation obtained using microfluidic CE-MS and LC-MS based peptide mapping suggests that the two methods generally generate very similar results (Fig. 4A, B). Furthermore, microfluidic CE-MS based peptide mapping demonstrates excellent repeatability as indicated by a tight standard deviation of six replicate injections.

Peptide mapping is a powerful characterization tool; however, several factors greatly limit application of peptide mapping. The first factor is a lengthy and labor-intensive sample preparation. The second factor is the long LC gradient time for LC-MS based peptide mapping. Since the digested samples are prone to artificial modification when they are stored in the autosampler, the number of samples that can be prepared and injected at one time are limited due to these artificial modifications. The third factor is the time-consuming and tedious data processing and reporting. Therefore, peptide mapping is normally only used for in-depth characterization and small-scale studies. Tremendous effort has been devoted to removing these obstacles, such as sample preparation automation [27] and data processing automation by multi-attribute methods (MAM) [4,28]. Microfluidic CE-MS peptide mapping method targets to significantly reduce instrument run time and improve the performance of MS. In addition, ZipChip is a nanoflow technique; therefore, it achieves superior sensitivity with significantly less amount of sample. Since it is a fast, low-sample-requirement, and highly reproducible method, it is suitable for monitoring predefined CQAs for large scale studies, such as design of experiments (DOEs) for clone selection, upstream and downstream process optimization, formulation development, stability and force degradation studies, etc. In contrast, conventional LC-MS methodology may be more suitable for in-depth characterization work, such as structure-function elucidation, comparability, and for products with little or no prior knowledge.

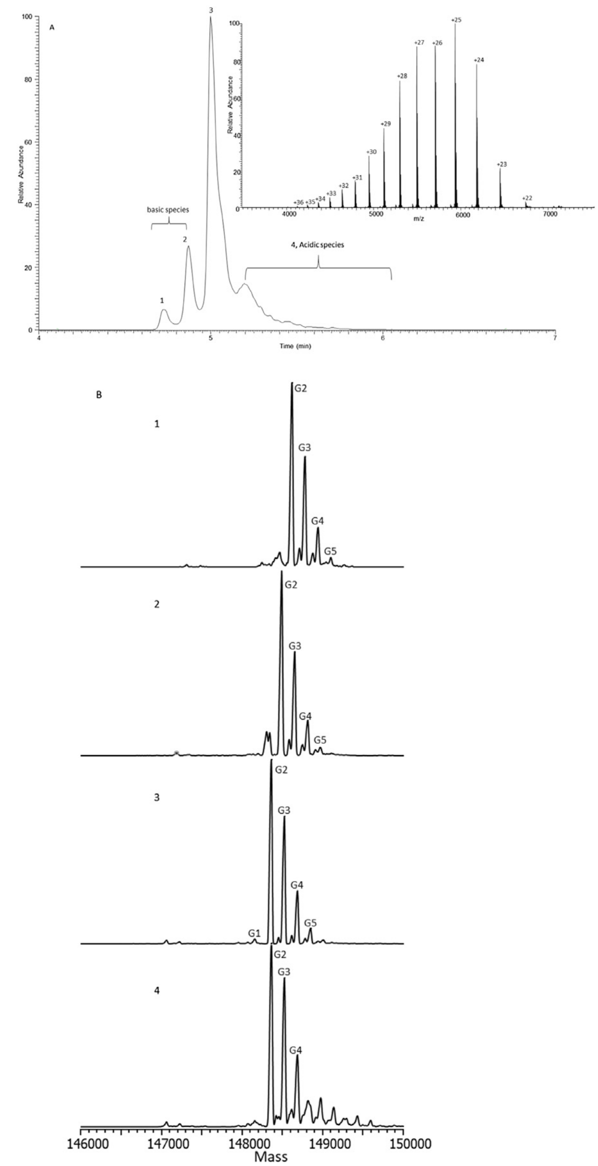

Fig. 5. A. Electropherogram of a GSK IgG1 mAb charge variants generated using microfluidic CE-MS based native intact analysis, and averaged raw mass spectrum of peak 3 (insert). B. Deconvoluted mass spectrum with peak identity for peak 1, 2, 3, and 4 shown in Fig. 6A.

3.5. *Native intact charge variants analysis using microfluidic CE-MS*

Charge variants of a GSK IgG1 and IgG4 mAb were characterized using microfluidic CE-MS based native intact analysis. The IgG1 mAb result demonstrated higher resolution than the IgG4 mAb due to its more basic pI. The representative IgG1 mAb electropherogram revealed four distinct product related peaks (Fig. 5A). The main peak (peak 3) migrated at ~5 min. Raw mass spectra of the main peak ranged from 4000 to 7000 m/z with charge states from +22 to +37 (insert of Fig. 5A), suggesting the protein remained in near native status when it was ionized. The peaks were labeled from 1 to 4 according to their migration time, and their raw mass spectra were deconvoluted respectively (Fig. 5B).

Major peaks in the deconvoluted mass spectra of peak 3 (main peak) were identified as glycoforms of the mAb with HC N-terminal Gln cyclized and HC C-terminal Lys clipped, according to their masses. Five major glycoforms (same or different glycans attached on both HCs) were assigned. The candidate glycoforms were selected based on microfluidic CE-MS based peptide mapping results of the mAb. The mass difference between glycoform G2 and G1 is 205 Da, suggesting that G2 had one more N-acetylglucosamine (GlcNAc, theoretical mass is 203 Da) than G1. The mass difference between glycoform G5, G4, G3, and G2 is 162 Da, suggesting that G3, G4, and G5 had one more hexose (theoretical mass is 162 Da) than G2, G3, and G4 respectively. The identified glycoforms and their relative abundance matched well with microfluidic CE-MS based peptide mapping (data not shown). The mass difference of glycoforms (G2, G3, G4, and G5) in peak 3 and glycoforms in peak 2 was 127 Da, and mass difference of glycoforms (G2, G3, G4, and G5) in peak 2 and in peak 1 was 129 Da. The 127 Da and 129 Da mass shift correspond to mass of Lys (theoretical Lys residual mass is 128 Da). Therefore, peak 1 contained glycoforms with HC N-terminal Gln cyclized and both HC C-terminal Lys remaining, while peak 2 contained glycoforms with HC N-terminal Gln cyclized and only one HC C-terminal Lys remaining. Hence, peak 1 and peak 2 were categorized as basic species. The mass difference of glycoforms in peak 3 and in peak 4 were 1 Da, which was within the instrumental mass accuracy. Since peak 4 migrated later than peak 3, masses in peak 4 were identified as acidic species, probably Asn deamidated forms due to 1 Da mass change. The acidic species may be consisted of several species without distinct separation. Further analysis is needed to better understand the acidic species.

3.6. Charge variants analysis by microfluidic CE-MS and cIEF

cIEF is the reference strategy for charge variant characterization in batch release tests. It is considered as a high resolution method with high reproducibility [29]. Therefore, a GSK IgG1 mAb was characterized using both microfluidic CE-MS platform and cIEF to compare the two methods. Both separations were visually comparable regarding overall profile and the number of separated features, except that migration order of basic and acidic species were reversed, as expected (Fig. 6A). The pH of ZipChip BGE for native intact analysis is 5.5. Basic species have a larger number of charges in acidic environment than main and acidic species, therefore they migrate faster in the microfluidic channel. In contrast, basic species have higher pI than main and acidic species, therefore it migrates slower in cIEF capillary. Both separations showed that the two basic species were distinctly separated with each other and with main species. The main species and acidic species were not clearly resolved. In addition, the acidic species peak was wide with tailing, which could likely represent a series of acidic species that are not well separated.

The relative abundance of basic, main, and acidic species determined using both methods were also compared (Fig. 6B). Results showed that the relative ratios of different charge variants determined by both methods were highly comparable. Furthermore, RSDs of charge variants relative abundance calculated using 6 replicates suggested that microfluidic CE-MS based native intact analysis had great reproducibility of less than 10%, which is also comparable to cIEF.

A common workflow to determine charge variants identity showed in a cIEF profile consists of multiple steps. First, develop a cation exchange (CEX) chromatography method that generates a charge variants profile similar to cIEF. Next, collect fractions containing individual charge variants. Third, perform peptide mapping analysis to determine charge variants identity. This workflow is laborious, time consuming, requires large amounts of sample and reagent, prone to contamination and human error. Recently, a pH gradient CEX-MS method was developed [30], hence charge variant separation and identification can be completed in a single analysis. However, the resolution of CEX separation normally is lower than cIEF, in addition, CEX-MS consumes more sample and requires more separation method development compared to microfluidic CE-MS based native intact analysis.

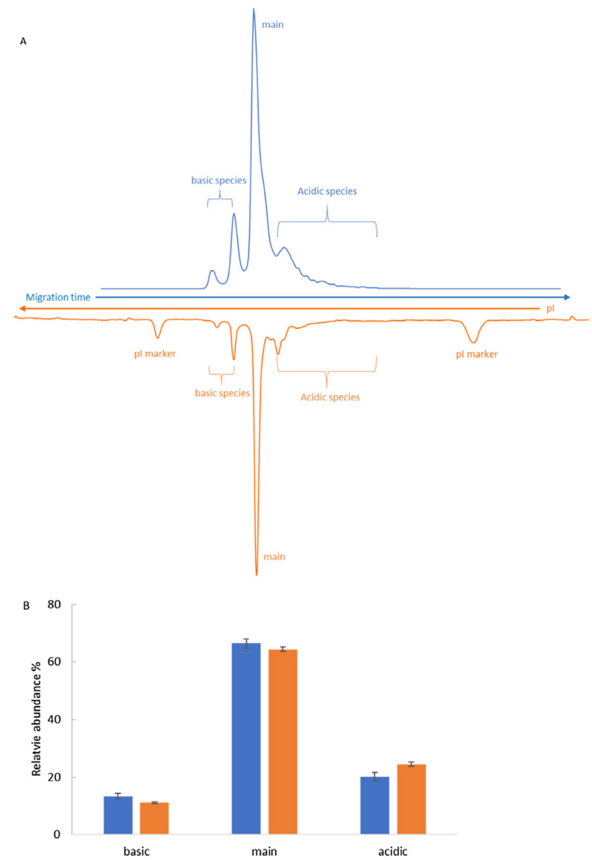

Fig. 6. A. Charge variant profiles of a GSK IgG 1 mAb determined using microfluidic CE-MS based native intact analysis (blue) and cIEF (orange). B. Relative abundance with standard deviation of basic, main, and acidic species determined using microfluidic CE-MS based native intact analysis (blue) and cIEF (orange).

3.7. Correlation between peptide mapping and native intact charge variants result

The GSK IgG1 mAb was characterized at both intact level and peptide level using the microfluidic CE-MS system. A reconstruction of an intact mass spectrum based on peptide mapping results was performed to evaluate the correlation between native intact data and peptide mapping data. The deconvoluted intact mass spectrum was derived from averaged mass spectra across all peaks shown in the native intact electropherogram from 4.6 min to 5.8 min (Fig. S4A). mAb major glycoforms and mAb with only one glycan attached were labelled on the deconvoluted spectra. Glycan species, various PTMs, including deamidation, oxidation, C-terminal and N-terminal variants etc., and their relative abundance determined by peptide mapping were imported into PMi Intact module to reconstruct the intact mass spectrum. The general profile of the reconstructed mass spectrum was comparable to experimentally determined intact mass spectra (Fig. SB). The model reconstruction displayed major glycoforms and mAb species with only one glycan that were showed in the experimental data. Visual inspection suggested that glycoform G1 peak was higher in the reconstructed spectrum, while G4 and G5 peaks were lower in the reconstructed spectrum compared with experimentally collected spectrum. Furthermore, mAb with only one glycan attached peaks were much higher in the simulated profile than in the observed profile. The difference between observed and simulated spectra can potentially be attributed to native intact and peptide mapping method variance, sample heterogeneity, and model algorithm. Nevertheless, model simulation demonstrated that peptide mapping data and native intact data were highly correlated. The model simulation made it possible to compare peptide mapping data to charge variant profile directly.

4. Conclusion

In this study, peptide mapping and native intact charge variants methods were developed using microfluidic CE-MS to conduct comprehensive characterization of mAbs. Compared with conventional LC-MS based methods, microfluidic CE-MS based peptide mapping monitors the same number of CQAs in a significantly shorter run time and lower sample consumption. Compared with industry standard cIEF, microfluidic CE-MS based native intact analysis provides a similar level of separation and identity of charge variant species. In addition, monitoring mAb PTMs, glycan profile at peptide level, and charge variants at intact level using one single system helps minimize instrument-introduced uncertainties and improve correlation between peptide mapping data and intact analysis.

The microfluidic CE-MS based peptide mapping and native intact analysis shows great potential in the biopharmaceutical product development process, especially in early product and process development stages, where design of experiments (DOEs) are extensively executed and limited material is available for analytical analysis. Once implemented, these methods will lead to accelerated biotherapeutic development timelines, an increased analytical analysis efficiency, and reduced development cost.

Authorship contribution statement

Li Cao: Conceptualization, Methodology – peptide mapping and model simulation, Data curation, Project administration, Writing – original draft, review, and editing. Daniel Fabry: Methodology – intact native charge variant method, Data curation, Writing – original draft, review, and editing. Kevin Lan: Conceptualization, Project administration, Writing – review and editing.

Declaration of Competing Interest

The authors report no declarations of interest.

Acknowledgments

The authors thank Troy Adams for leading the technology evaluation activity, Dr. Hillary Schuessler, Dr. Aming Zhang, and Lee Oliver for critical review of the manuscript. The authors also thank Dr. Aston Liu for providing support throughout the project. The authors also acknowledge 908 Devices and Protein Metrics for their collaboration and customer support.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.jpba.2021.114251.

References

[1] A. Mullard, FDA approves 100th monoclonal antibody product, Nat Rev Drug Discov (2021).
[2] Monoclonal Antibodies (MAbS) Global Market Report 2021: COVID 19 Impact And Recovery To 2030, The business research company, January 2021, 2021.
[3] L.X. Yu, G. Amidon, M.A. Khan, S.W. Hoag, J. Polli, G.K. Raju, J. Woodcock, Understanding pharmaceutical quality by design, AAPS J 16 (4) (2014) 771–783.
[4] R.S. Rogers, N.S. Nightlinger, B. Livingston, P. Campbell, R. Bailey, A. Balland, Development of a quantitative mass spectrometry multi-attribute method for characterization, quality control testing and disposition of biologics, MAbs 7 (5) (2015) 881–890.
[5] M.J. Badgett, B. Boyes, R. Orlando, The Separation and Quantitation of Peptides with and without Oxidation of Methionine and Deamidation of Asparagine Using Hydrophilic Interaction Liquid Chromatography with Mass Spectrometry (HILIC-MS), J Am Soc Mass Spectrom 28 (5) (2017) 818–826.
[6] F. Kilar, Recent applications of capillary isoelectric focusing, Electrophoresis 24 (22-23) (2003) 3908–3916.
[7] S. Fekete, A. Beck, J. Fekete, D. Guillarme, Method development for the separation of monoclonal antibody charge variants in cation exchange chromatography, Part I: salt gradient approach, J Pharm Biomed Anal 102 (2015) 33–44.
[8] Y. He, C. Isele, W. Hou, M. Ruesch, Rapid analysis of charge variants of monoclonal antibodies with capillary zone electrophoresis in dynamically coated fused-silica capillary, J Sep Sci 34 (5) (2011) 548–555.
[9] E. Farsang, D. Guillarme, J.L. Veuthey, A. Beck, M. Lauber, A. Schmudlach, S. Fekete, Coupling non-denaturing chromatography to mass spectrometry for the characterization of monoclonal antibodies and related products, J Pharm Biomed Anal 185 (2020), 113207.
[10] A. Stolz, K. Jooss, O. Hocker, J. Romer, J. Schlecht, C. Neususs, Recent advances in capillary electrophoresis-mass spectrometry: Instrumentation, methodology and applications, Electrophoresis 40 (1) (2019) 79–112.
[11] S. Mack, D. Arnold, G. Bogdan, L. Bousse, L. Danan, V. Dolnik, M. Ducusin, E. Gwerder, C. Herring, M. Jensen, J. Ji, S. Lacy, C. Richter, I. Walton, E. Gentalen, A novel microchip-based imaged CIEF-MS system for comprehensive characterization and identification of biopharmaceutical charge variants, Electrophoresis 40 (23-24) (2019) 3084–3091.
[12] J.S. Mellors, V. Gorbounov, R.S. Ramsey, J.M. Ramsey, Fully integrated glass microfluidic device for performing high-efficiency capillary electrophoresis and electrospray ionization mass spectrometry, Anal Chem 80 (18) (2008) 6881–6887.
[13] E.A. Redman, N.G. Batz, J.S. Mellors, J.M. Ramsey, Integrated microfluidic capillary electrophoresis-electrospray ionization devices with online MS detection for the separation and characterization of intact monoclonal antibody variants, Anal Chem 87 (4) (2015) 2264–2272.
[14] S. Carillo, C. Jakes, J. Bones, In-depth analysis of monoclonal antibodies using microfluidic capillary electrophoresis and native mass spectrometry, J Pharm Biomed Anal 185 (2020), 113218.
[15] F. Füssl, A. Trappe, S. Carillo, C. Jakes, J. Bones, Comparative Elucidation of Cetuximab Heterogeneity on the Intact Protein Level by Cation Exchange Chromatography and Capillary Electrophoresis Coupled to Mass Spectrometry, Analytical Chemistry 92 (7) (2020) 5431–5438.
[16] E.A. Redman, J.S. Mellors, J.A. Starkey, J.M. Ramsey, Characterization of Intact Antibody Drug Conjugate Variants Using Microfluidic Capillary Electrophoresis-Mass Spectrometry, Anal Chem 88 (4) (2016) 2220–2226.
[17] A.B. Dykstra, T.G. Flick, B. Lee, L.E. Blue, N. Angell, Chip-Based Capillary Zone Electrophoresis Mass Spectrometry for Rapid Resolution and Quantitation of Critical Quality Attributes in Protein Biotherapeutics, J Am Soc Mass Spectrom (2021).
[18] K. Khatri, J.A. Klein, J.R. Haserick, D.R. Leon, C.E. Costello, M.E. McComb, J. Zaia, Microfluidic Capillary Electrophoresis-Mass Spectrometry for Analysis of Monosaccharides, Oligosaccharides, and Glycopeptides, Anal Chem 89 (12) (2017) 6645–6655.
[19] Y. Liu, J. Fernandez, Z. Pu, H. Zhang, L. Cao, I. Aguilar, D. Ritz, R. Luo, A. Read, A. Laures, K. Lan, A. Ubiera, A. Smith, P. Patel, A. Liu, Simultaneous Monitoring and Comparison of Multiple Product Quality Attributes for Cell Culture Processes at Different Scales Using a LC/MS/MS Based Multi-Attribute Method, J Pharm Sci 109 (11) (2020) 3319–3329.
[20] M. Bern, Y.J. Kil, C. Becker, Byonic: advanced peptide and protein identification software, Curr Protoc Bioinformatics Chapter 13, 2012, Unit13 20.
[21] H. Yang, R.A. Zubarev, Mass spectrometric analysis of asparagine deamidation and aspartate isomerization in polypeptides, Electrophoresis 31 (11) (2010) 1764–1772.
[22] P. Hao, J. Qian, B. Dutta, E.S. Cheow, K.H. Sim, W. Meng, S.S. Adav, A. Alpert, S.K. Sze, Enhanced separation and characterization of deamidated peptides with RP-ERLIC-based multidimensional chromatography coupled with tandem mass spectrometry, J Proteome Res 11 (3) (2012) 1804–1811.
[23] K. Morand, G. Talbo, M. Mann, Oxidation of peptides during electrospray ionization, Rapid Commun Mass Spectrom 7 (8) (1993) 738–743.
[24] B.K. Mautz, Larraillet Maximiliane, Vincent Mølhøj, Michael Monitoring of On-column Methionine Oxidation as Part of a System Suitability Test During UHPLC–MS/MS Peptide Mapping, LCGC 37 (11) (2019) 8–13.
[25] G.S.M. Lageveen-Kammeijer, N. de Haan, P. Mohaupt, S. Wagt, M. Filius, J. Nouta, D. Falck, M. Wuhrer, Highly sensitive CE-ESI-MS analysis of N-glycans from complex biological samples, Nat Commun 10 (1) (2019) 2137.
[26] J. Zaia, Capillary electrophoresis-mass spectrometry of carbohydrates, Methods Mol Biol 984 (2013) 13–25.
[27] J. Richardson, B. Shah, G. Xiao, P.V. Bondarenko, Z. Zhang, Automated in-solution protein digestion using a commonly available high-performance liquid chromatography autosampler, Anal Biochem 411 (2) (2011) 284–291.

[28] Y. Wang, X. Li, Y.H. Liu, D. Richardson, H. Li, M. Shameem, X. Yang, Simultaneous monitoring of oxidation, deamidation, isomerization, and glycosylation of monoclonal antibodies by liquid chromatography-mass spectrometry method with ultrafast tryptic digestion, MAbs 8 (8) (2016) 1477–1486.

[29] J. Kahle, H. Zagst, R. Wiesner, H. Watzig, Comparative charge-based separation study with various capillary electrophoresis (CE) modes and cation exchange chromatography (CEX) for the analysis of monoclonal antibodies, J Pharm Biomed Anal 174 (2019) 460–470.

[30] F. Fussl, K. Cook, K. Scheffler, A. Farrell, S. Mittermayr, J. Bones, Charge Variant Analysis of Monoclonal Antibodies Using Direct Coupled pH Gradient Cation Exchange Chromatography to High-Resolution Native Mass Spectrometry, Anal Chem 90 (7) (2018) 4669–4676.